

JavaScript Array Method - push()

The `push()` method is one of the most commonly used array methods in JavaScript. It is used to add one or more elements to the **end** of an array.

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What is push()?

Definition

The `push()` method adds one or more elements to the **end** of an array.

It modifies the **original array** and returns the **new length** of the array.

Why Use push()?

We use `push()` when we need to:

- Add a new item to an array
 - Store user input
 - Add products to a shopping cart
 - Save messages in a chat application
 - Store API data
 - Maintain dynamic lists
-

Syntax

```
array.push(element1);
```

```
array.push(element1, element2, element3);
```

Parameters

Parameter	Description
element1	First element to add
element2	Second element (Optional)
element3	Third element (Optional)

You can add **one or many elements** in a single call.

Return Value

`push()` returns the **new length** of the array.

Example

```
const fruits = ["Apple", "Mango"];

let length = fruits.push("Orange");

console.log(length);
```

Output

```
3
```

How push() Works

Before

```
["Apple", "Mango"]
```

↓

```
push("Orange")
```

↓

After

```
["Apple", "Mango", "Orange"]
```

The new element is always added to the **end** of the array.

Example 1 (Basic)

```
const fruits = [  
  
  "Apple",  
  
  "Mango"  
  
];  
  
fruits.push("Orange");  
  
console.log(fruits);
```

Output

```
["Apple", "Mango", "Orange"]
```

Explanation

- Original array contains two elements.
 - `push("Orange")` adds `"Orange"` at the end.
 - The original array is modified.
-

Example 2 (Numbers)

```
const numbers = [  
  
  10,  
  
  20,  
  
  30  
  
];  
  
numbers.push(40);  
  
console.log(numbers);
```

Output

```
[10,20,30,40]
```

Example 3 (Multiple Elements)

```
const colors = [  
  "Red",  
  "Blue"  
];  
  
colors.push(  
  "Green",  
  "Black",  
  "White"  
);  
  
console.log(colors);
```

Output

```
["Red", "Blue", "Green", "Black", "White"]
```

Example 4 (Store Return Value)

```
const fruits = [  
  "Apple",  
  "Mango"  
];  
  
let newLength = fruits.push("Orange");  
  
console.log(newLength);  
  
console.log(fruits);
```

Output

```
3  
  
["Apple","Mango","Orange"]
```

Example 5 (Objects)

```
const users = [  
  
  {  
  
    name: "John"  
  
  }  
  
];  
  
users.push({  
  
  name: "Sara"  
  
});  
  
console.log(users);
```

Output

```
[  
  { name: "John" },  
  { name: "Sara" }  
]
```

Example 6 (Booleans)

```
const values = [  
  true,  
  false  
];  
  
values.push(true);  
  
console.log(values);
```

Output

```
[true,false,true]
```

Example 7 (Mixed Data Types)

```
const data = [  
  "Rokon",  
  23  
];  
  
data.push(  
  true,  
  null  
);  
  
console.log(data);
```

Output

```
["Rokon",23,true,null]
```

Example 8 (Empty Array)

```
const numbers = [];  
  
numbers.push(100);  
  
console.log(numbers);
```

Output

```
[100]
```

Real Project Example 1

Shopping Cart

```
const cart = [  
  "Laptop",  
  "Mouse"  
];  
  
cart.push("Keyboard");  
  
console.log(cart);
```

Output

```
["Laptop", "Mouse", "Keyboard"]
```

Real Project Example 2

Student List

```
const students = [  
  "Rahim",  
  "Karim"  
];  
  
students.push("Rokon");  
  
console.log(students);
```

Output

```
["Rahim", "Karim", "Rokon"]
```

Real Project Example 3

Notifications

```
const notifications = [];  
  
notifications.push("New Message");  
  
notifications.push("Payment Successful");  
  
console.log(notifications);
```

Output

```
[  
  "New Message",  
  "Payment Successful"  
]
```

Real Project Example 4

Chat Application

```
const messages = [];  
  
messages.push("Hello");  
  
messages.push("How are you?");  
  
console.log(messages);
```

Output

```
[  
  "Hello",  
  "How are you?"  
]
```

Real Project Example 5

Todo List

```
const todos = [  
  "Study"  
];  
  
todos.push("Exercise");  
  
todos.push("Read Book");  
  
console.log(todos);
```

Output

```
[  
  "Study",  
  "Exercise",  
  "Read Book"  
]
```

Best Practices

- ✓ Use `push()` to add elements at the end of an array.

```
cart.push(product);
```

- ✓ Add multiple elements in one call when appropriate.

```
numbers.push(4, 5, 6);
```

- ✓ Store the return value if you need the updated array length.

```
const length = numbers.push(100);
```

- ✓ Use meaningful array names.

```
students
```

```
products
```

```
orders
```

```
messages
```

Common Mistakes

Mistake 1

Expecting `push()` to return the array.

Wrong

```
const result = numbers.push(4);  
  
console.log(result);
```

Output

```
4
```

`push()` returns the **new length**, not the array.

Correct

```
numbers.push(4);  
  
console.log(numbers);
```

Mistake 2

Using `push()` on a string.

Wrong

```
let name = "John";  
  
name.push("A");
```

Output

TypeError

`push()` works only with arrays.

Mistake 3

Forgetting that `push()` changes the original array.

```
const numbers = [1,2];  
  
numbers.push(3);
```

Now

```
numbers
```

becomes

```
[1,2,3]
```

Mistake 4

Using the wrong method.

To add at the **beginning**, use

```
unshift()
```

not

```
push()
```

Interview Questions

What does `push()` do?

It adds one or more elements to the **end** of an array.

Does `push()` modify the original array?

✓ Yes.

What does `push()` return?

It returns the **new length** of the array.

Can `push()` add multiple elements?

✓ Yes.

```
numbers.push(4,5,6);
```

Which end of the array does `push()` add elements to?

The **end** of the array.

Which method removes the last element?

`pop()`

Which method adds an element at the beginning?

`unshift()`

Summary Table

Feature	Description
Method	<code>push()</code>
Purpose	Add element(s) to the end of an array
Parameters	One or more elements
Returns	New array length
Changes Original Array	☒ Yes
Adds To	End of the array
Common Uses	Shopping Cart, Todo List, Chat Messages, Notifications

Quick Comparison

Method	Action	Returns	Changes Original Array
<code>push()</code>	Add to end	New length	☒ Yes
<code>pop()</code>	Remove from end	Removed element	☒ Yes
<code>unshift()</code>	Add to beginning	New length	☒ Yes
<code>shift()</code>	Remove from beginning	Removed element	☒ Yes

Final Notes

`push()` is one of the most frequently used array methods in JavaScript. It is widely used in:

- ☒ React state updates (when creating new arrays)
- ☒ Shopping carts
- ☒ Todo applications
- ☒ Chat systems
- ☒ Notifications
- ☒ API data processing
- ☒ Dynamic lists

Mastering `push()` is essential before learning other array methods like `pop()`, `shift()`, `unshift()`, `splice()`, and `concat()`.